

CODE SPLASH

HACKATHON PHASE

BestRoute

Competition Theme_05

CodeCrushers

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1. EXECUTIVE SUMMARY

Public transportation journeys often require passengers to combine multiple transportation modes such as buses, trains, taxis, three-wheelers, and walking. In many cases, passengers must plan each part of the journey separately, making it difficult to compare complete routes, estimate total costs, manage transfers, and respond to delays or cancellations [2], [6]. This can result in longer waiting times, missed connections, increased travel costs, and uncertainty during journeys.

BestRoute is an intelligent multimodal public transportation optimization platform designed to provide passengers with a unified solution for planning, comparing, and monitoring complete journeys. Instead of treating each transportation mode as a separate service, BestRoute considers the entire journey as a connected route and identifies suitable combinations of transportation modes.

Passengers can enter their origin, destination, preferred travel time, and journey preferences. The system generates possible multimodal routes and evaluates them using factors such as travel time, waiting time, cost, walking distance, number of transfers, service reliability, and connection risk. Routes are then ranked according to the passenger's selected preference, such as fastest, cheapest, minimum walking, or minimum transfers.

A key feature of BestRoute is its ability to respond to transportation disruptions. When a delay, cancellation, road closure, or other disruption affects a planned journey, the system can identify alternative routes and provide an updated recommendation.

The primary users are daily commuters, students, tourists, and other public transportation passengers. Transportation operators and authorities can also benefit from transportation data and usage insights. The expected impact includes improved journey planning, reduced waiting time and transfer uncertainty, and more efficient use of available transportation services.

The proposed system will consist of a passenger mobile application, backend services, transportation data layer, multimodal route optimization engine, and administrative dashboard. The main technologies include React Native, React, Node.js, Express.js, MongoDB, mapping and geospatial services, and multi-criteria optimization algorithms.

2. PROBLEM STATEMENT

2.1 Problem Description

Passengers travelling through multiple transportation modes often need to plan each part of their journey independently. A journey may require a bus to a railway station, a train to another location, and a taxi or three-wheeler to reach the final destination [2].

This fragmented planning process creates several problems:

- Difficulty identifying suitable combinations of transportation modes.
- Difficulty comparing complete journey duration and cost.
- Uncertainty regarding transfer times.
- Increased risk of missed connections.
- Limited support when services are delayed or cancelled.
- Lack of a unified view of the complete journey.

As a result, passengers may select transportation options based on incomplete information rather than the most suitable combination for their needs.

2.2 Background and Context

Public transportation networks consist of multiple operators, routes, schedules, fares, and service conditions. Although individual transportation services may provide information about their own operations, passengers often need to coordinate information from several services to complete one journey.

For example:

Bus → Train → Taxi

requires the passenger to determine:

1. Which bus to take.
2. Where to transfer to the train.
3. Whether sufficient transfer time is available.
4. Which train is suitable?
5. How to complete the final part of the journey.
6. The total travel time and cost.

The problem becomes more significant when one service is delayed. A delay in the first segment can make the remaining planned journey unsuitable.

IntelliTransit addresses this issue by considering the complete journey rather than treating each transportation segment independently.

2.3 Importance of the Problem

Effective multimodal journey planning can improve the passenger experience by considering multiple factors rather than travel time alone.

The proposed solution aims to contribute to:

- Reduced waiting time.
- Fewer missed connections.
- Better route selection.
- Improved travel convenience.
- Greater visibility of transportation alternatives.
- More informed use of public transportation.
- Reduced uncertainty during multimodal journeys.

The problem is therefore relevant to both individual passengers and transportation stakeholders.

3. PROPOSED TECHNICAL SOLUTION

3.1 Solution Overview

BestRoute will provide a unified platform for planning, comparing, and monitoring multimodal public transportation journeys.

The passenger will provide:

- Origin.
- Destination.
- Desired departure or arrival time.
- Journey preferences.

The system will identify available transportation options and generate possible multimodal journeys. Each journey will be evaluated using multiple criteria.

Evaluation Factor	Purpose
Travel Time	Estimate the total journey duration
Waiting Time	Identify time spent waiting between services
Cost	Estimate the total journey cost
Transfers	Measure the number of transportation changes
Walking Distance	Estimate walking required during the journey
Reliability	Consider expected service reliability
Connection Risk	Identify potentially difficult transfers

Table 3.1: Multimodal Journey Evaluation Factors

The optimization engine will rank the generated routes according to the passenger's preferences. If the selected journey is affected by a disruption, the system will generate alternative routes using the remaining available transportation services.

Example:

Journey: Kandy → Colombo

Preference: Fastest + Affordable

Possible recommendation:

Bus → Train → Taxi

The system can display:

- Estimated total duration.
- Estimated total cost.
- Departure and arrival times.
- Transfer locations.
- Waiting time.
- Walking distance.
- Connection risk.

If the selected train is delayed, BestRoute can identify an alternative journey and notify the passenger.

3.2 Key Features

3.2.1 Unified Multimodal Journey Planner

The platform will combine multiple transportation modes into a single journey-planning process. The initial prototype may support:

- Buses.
- Trains.
- Taxis.
- Three-wheelers.
- Walking.

The architecture will allow additional transportation modes to be integrated in the future.

3.2.2 Personalized Journey Optimization

Passengers can select their preferred journey criteria:

- Fastest.
- Cheapest.
- Minimum walking.
- Minimum transfers.
- Most reliable.

The route-ranking process will adjust according to the selected preference [3].

3.2.3 Complete Journey Comparison

The system will present multiple route options and allow passengers to compare:

- Total travel time.
- Estimated cost.
- Waiting time.
- Number of transfers.
- Walking distance.
- Reliability.
- Connection risk.

3.2.4 Connection-Risk Detection

The system will evaluate whether sufficient time is available between connecting services.

For example:

Event	Time
Train Arrival	10:15 AM
Bus Departure	10:20 AM
Available Transfer Time	5 minutes

Table 3.2: Connection Risk Evaluation Example

If the estimated transfer requirement exceeds the available time, the connection will be identified as high risk.

3.2.5 Disruption Monitoring

The system will identify available transportation disruptions such as:

- Service delays.
- Cancellations.
- Road closures.
- Traffic-related interruptions.
- Route interruptions.

3.2.6 Dynamic Journey Re-optimization

When a disruption affects an active journey, the system will recalculate the remaining journey and recommend an alternative.

Original:

Bus → Train → Taxi

Alternative:

Bus → Train → Bus → Taxi

The passenger will receive the updated recommendation through the application.

3.2.7 Unified Journey Cost

The system will calculate the estimated total cost of the complete journey rather than requiring passengers to calculate individual transportation costs separately.

3.2.8 Passenger Notifications

Notifications will inform passengers about:

- Upcoming transfers.
- Delays.
- Connection risks.
- Journey changes.
- Alternative routes.

3.2.9 Administrative Dashboard

The administrative dashboard will support:

- Transportation service management.
- Route and schedule management.
- Disruption management.
- Transportation data monitoring.
- Basic usage analytics.

3.3 Innovation and Uniqueness

The main innovation of BestRoute is its focus on the complete multimodal journey rather than individual transportation services. The solution combines four key capabilities:

Capability	Description
Multimodal Integration	Combines different transportation modes into one journey
Personalized Optimization	Ranks routes according to passenger preferences
Connection-Risk Evaluation	Evaluates whether transfers are realistically achievable
Dynamic Re-optimization	Updates the journey when transportation conditions change

Table 3.3: Key Innovations and Unique Capabilities

This approach allows IntelliTransit to function as a journey decision-support and optimization platform, rather than simply providing transportation information.

3.4 System Architecture

The proposed system will use a layered architecture.

Presentation Layer

- Passenger mobile application.
- Administrative web dashboard.

Application Layer

- Authentication and authorization.
- Journey planning service.
- Route management.
- Notification service.
- User preference management.

Optimization Layer

- Multimodal route generation.
- Route scoring.
- Connection-risk evaluation.
- Disruption handling.
- Dynamic re-routing.

Transportation Data Layer

- Transportation services.
- Routes.
- Stops and stations.
- Schedules.
- Travel times.
- Fares.
- Service status.

Database Layer

The database will store:

- User information.
- Transportation data.
- Journey requests.
- Journey history.
- Disruptions.
- Preferences.
- Notifications.

External Services

Where required, the system may integrate with:

- Mapping services.
- Geolocation services.
- Transportation APIs.
- Push notification services.

Architecture Flow:

Passenger Application → Backend API → Optimization Engine → Transportation Data /
External Services → Route Recommendation → Passenger

3.5 Data Flow

The primary journey-planning process will follow these steps:

1. Passenger enters the origin and destination.
2. Passenger selects journey preferences.
3. The system obtains relevant location information.
4. Nearby transportation stops and services are identified.
5. Transportation schedules and availability are retrieved.
6. Possible multimodal journeys are generated.
7. Candidate journeys are evaluated.
8. Routes are ranked according to passenger preferences.
9. Recommended routes are presented to the passenger.
10. The passenger selects and starts a journey.
11. The journey is monitored for relevant disruptions.
12. If necessary, an alternative journey is generated.

3.6 Technology Justification

Technology	Purpose	Justification
React Native	Passenger mobile application	Supports cross-platform mobile development and map-based interfaces
React	Administrative dashboard	Suitable for interactive web-based management interfaces
Node.js	Backend runtime	Supports scalable API and real-time application services

Express.js	Backend framework	Provides a lightweight structure for REST APIs
MongoDB	Database	Flexible document structure suitable for transportation and user data
Leaflet / Mapping API	Maps and routes	Supports geographical visualization and route-related features
JWT	Authentication	Supports secure token-based authentication
Git/GitHub	Version control	Enables collaborative development and version management
Figma	UI/UX design	Supports interface design and prototyping
Postman	API testing	Supports development and validation of backend APIs

Table 3.4: Technology Justification

3.7 AI and Optimization Approach

The core prototype will use a multi-criteria route optimization approach rather than depending on a complex machine-learning model.

A route can be evaluated using:

- Travel time.
- Cost.
- Waiting time.
- Number of transfers.
- Walking distance.
- Reliability.
- Connection risk.

A simplified scoring model is:

$$\text{Route Score} = w_1(\text{Time}) + w_2(\text{Cost}) + w_3(\text{Waiting}) + w_4(\text{Transfers}) + w_5(\text{Walking}) + w_6(\text{Risk})$$

The weights will change according to passenger preferences.

For example:

Preference	Higher Weight
Fastest	Travel time
Cheapest	Cost
Minimum Walking	Walking distance
Minimum Transfers	Number of transfers
Most Reliable	Reliability and connection risk

Table 3.4: Passenger Preference Weighting Criteria

If sufficient historical data is available, machine-learning techniques may be incorporated for:

- Travel-time prediction.
- Delay prediction.
- Connection-risk prediction.

The final implementation will prioritize approaches that can be realistically developed, evaluated, and demonstrated within the competition period.

3.8 Security Considerations

Security will be incorporated into the system design through:

- Secure authentication.
- Password hashing.
- Role-based access control.
- Protected API endpoints.
- Input validation.
- HTTPS communication.
- Controlled administrative access.
- Secure handling of user location information.

Location information will only be collected and processed when required for journey planning or related functionality.

4. TARGET AUDIENCE

4.1 Primary Users

User Group	Requirement
Daily Commuters	Efficient and reliable multimodal journeys
Students	Affordable transportation combinations
Tourists and Visitors	Simple guidance through unfamiliar transportation networks
Occasional Passengers	Assistance in selecting suitable transportation combinations

Table 4.1: Target User Groups and Their Requirements

4.2 Beneficiaries

The main beneficiaries are:

- Public transportation passengers.
- Students.
- Daily commuters.
- Tourists.
- Transportation operators.
- Transport authorities.
- Urban planners.

Passengers benefit from improved journey planning, while transportation stakeholders can gain useful aggregated insights into transportation usage.

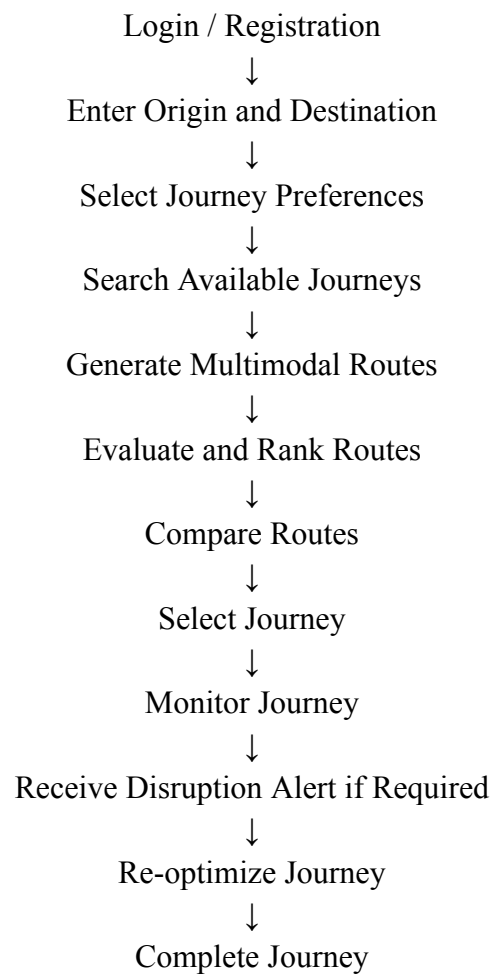
4.3 Stakeholders

Potential stakeholders include:

- Bus service providers.
- Railway service providers.
- Taxi and ride-hailing providers.
- Transportation authorities.
- Municipal authorities.
- Passengers.
- Tourism organizations.
- Urban planning organizations.

4.4 User Flow

The main passenger flow will be:



The detailed user flow diagram will be included in Section 8.

4.5 Key Use Cases

ID	Use Case	Actor	Priority
UC01	Register and Authenticate	Passenger / Admin	High
UC02	Manage Travel Preferences	Passenger	High
UC03	Search Multimodal Journey	Passenger	High
UC04	Generate Multimodal Routes	System	High
UC05	Score and Rank Routes	System	High
UC06	Evaluate Connection Risk	System	High
UC07	Compare and Select Route	Passenger	High
UC08	Monitor Active Journey	System	High
UC09	Detect and Notify Disruptions	System / Admin	Very High
UC10	Re-optimize Journey	System	Very High
UC11	Manage Routes and Schedules	Admin	High
UC12	Manage Service Disruptions	Admin	Very High
UC13	View Transportation Analytics	Admin	Medium
UC14	Save Favourite Journeys	Passenger	Medium

Table 4.2: Key Use Cases

5. TECHNICAL AND BUSINESS FEASIBILITY

5.1 Technical Feasibility

The proposed system is technically feasible because its core components can be developed using established mobile, web, backend, database, mapping, and optimization technologies. The initial prototype will focus on a controlled geographical area and selected transportation services rather than attempting to integrate every available transportation provider. The system can be developed using:

- Passenger mobile application.
- Backend API.
- MongoDB database.
- Transportation dataset.
- Mapping and geolocation services.
- Route optimization engine.
- Administrative dashboard.

This approach keeps the prototype achievable within the competition timeline while demonstrating the core concept.

5.2 Resource Requirements

The development of BestRoute will require a combination of hardware, software, transportation data, development tools, and external services. The required resources are summarized below.

Resource Category	Required Resources	Purpose
Hardware	Developer computers/laptops	Software development and testing
Hardware	GPS-enabled smartphones	Mobile application and location testing
Software	React Native	Passenger mobile application development
Software	React	Administrative dashboard development
Software	Node.js and Express.js	Backend API and application services
Software	MongoDB	Storage of users, routes, schedules, journeys, and other system data
Development Tools	VS Code, Git/GitHub, Postman	Development, version control, and API testing

Design Tools	Figma	UI/UX design and prototyping
Transportation Data	Bus and train routes, stops, schedules, fares, and travel times	Multimodal journey planning and route optimization
Location Services	GPS and mapping/geolocation services	Location detection, nearby stops, and route visualization
External Services	Transportation APIs or publicly available datasets	Obtaining transportation information
Notification Services	Push notification service	Delay, disruption, transfer, and journey notifications
Human Resources	Development and testing team members	System development, integration, testing, and documentation
Hosting	Cloud-based backend/database services	Deployment and demonstration of the prototype

Table 5.1: Resource Requirements

For the initial prototype, the system will focus on a limited geographical area and selected transportation services. This approach reduces the amount of data and infrastructure required while allowing the core multimodal journey-planning and optimization features to be demonstrated effectively.

5.3 Potential Challenges and Mitigation

Challenge	Mitigation
Limited transportation APIs	Use controlled/public datasets and design an extensible integration layer
Limited real-time data	Use simulated disruptions for the prototype while keeping the architecture API-ready
Large number of possible routes	Filter routes using distance, time, availability, and transfer constraints
Inaccurate transportation data	Validate datasets and clearly identify estimated information
Increasing system complexity	Use modular transportation-provider integrations
Limited historical data for ML	Prioritize rule-based optimization and use ML only where sufficient data exists

Table 5.2: Potential Challenges and Mitigation

6. DEVELOPMENT ROADMAP

6.1 Development Plan

Phase	Description	Main Activities
1	Requirements and Research	Define problem, users, transportation modes, requirements, and datasets
2	UI/UX and Database Design	Design passenger interface, dashboard, database, and transportation data model
3	Backend Development	Develop authentication, APIs, transportation services, and notifications
4	Route Optimization	Implement route generation, route scoring, preferences, and connection-risk evaluation
5	Mobile Application	Implement journey search, maps, route comparison, tracking, and notifications
6	Intelligent Features	Implement disruption detection and dynamic re-routing
7	Testing	Conduct functional, API, performance, security, route accuracy, and usability testing
8	Final Demonstration	Prepare documentation, presentation, demonstration scenario, and final pitch

Table 6.1: Development Plan

6.2 Technology Stack

Category	Proposed Technology
Mobile Application	React Native
Web Dashboard	React
Backend	Node.js + Express.js
Database	MongoDB
Maps	Leaflet / Mapping API
Geolocation	Device GPS / Geolocation API
Authentication	JWT
API Testing	Postman
Version Control	Git / GitHub
UI/UX	Figma
Optimization	Multi-criteria route optimization
Optional ML	Python
Notifications	Push Notification Service
Hosting	Cloud-based deployment platform

Table 6.2: Technology Stack

6.3 Key Milestones

Milestone	Expected Outcome
Requirements Completed	Defined system requirements
UI/UX Completed	Passenger and admin designs
Database Completed	Transportation and user data models
Backend Completed	Core APIs and services
Route Engine Completed	Multimodal route generation
Optimization Completed	Personalized route ranking
Mobile Application Completed	Functional passenger application
Disruption Module Completed	Dynamic alternative journey generation
Testing Completed	Validated integrated prototype
Final Demonstration	Demonstration-ready system

Table 6.3: Key Milestones

6.4 Expected Deliverables

The project is expected to deliver:

1. Passenger mobile application.
2. Administrative web dashboard.
3. Multimodal journey-planning engine.
4. Transportation data management module.
5. Personalized route optimization module.
6. Connection-risk evaluation module.
7. Dynamic re-routing mechanism.
8. Notification system.
9. Backend APIs.
10. MongoDB database.
11. Technical documentation.
12. Demonstration-ready prototype.

6.5 Potential Technical Challenges

The main technical challenges are:

- Integrating different transportation datasets.
- Handling inconsistent schedules.
- Generating efficient multimodal routes.
- Processing transportation disruptions.
- Maintaining route-calculation performance.
- Maintaining accurate geographical information.
- Scaling to additional transportation providers.
- Applying prediction models when historical data is limited.

These challenges will be addressed through modular architecture, controlled datasets, route filtering, validation, caching, and incremental development.

7. SUSTAINABILITY AND FUTURE ENHANCEMENTS

7.1 Long-Term Viability

BestRoute can be extended from a competition prototype into a broader multimodal mobility platform.

The modular architecture will allow additional transportation providers, geographical regions, and services to be introduced without redesigning the complete system.

Potential applications include:

- City-level transportation planning.
- Regional multimodal transportation.
- Tourism transportation.
- Smart-city mobility management.
- Transportation demand analysis.

7.2 Scalability

The system can be scaled through:

- Modular transportation-provider integrations.
- RESTful APIs.
- Cloud deployment.
- Database indexing.
- Route-search optimization.
- Caching.
- Independent system components.

The platform can initially operate within a limited geographical area and gradually expand to additional cities and transportation networks.

7.3 Future Improvements

Future versions may introduce:

- Predictive Transportation Analytics
Predict travel demand, delays, and journey duration using historical transportation data.
- Smart Ticketing
Support ticket management across participating transportation services.
- Integrated Digital Payments
Enable unified payments for supported transportation providers.
- Accessibility-Aware Routing
Provide routes optimized for passengers with different mobility requirements.
- Carbon-Aware Journey Planning
Allow passengers to compare journeys based on estimated environmental impact.
- Provider Analytics
Provide transportation operators with anonymized insights into demand patterns and passenger preferences.

8. SKETCHES, DIAGRAMS, MOCKUPS AND WIREFRAMES

8.1 System Architecture

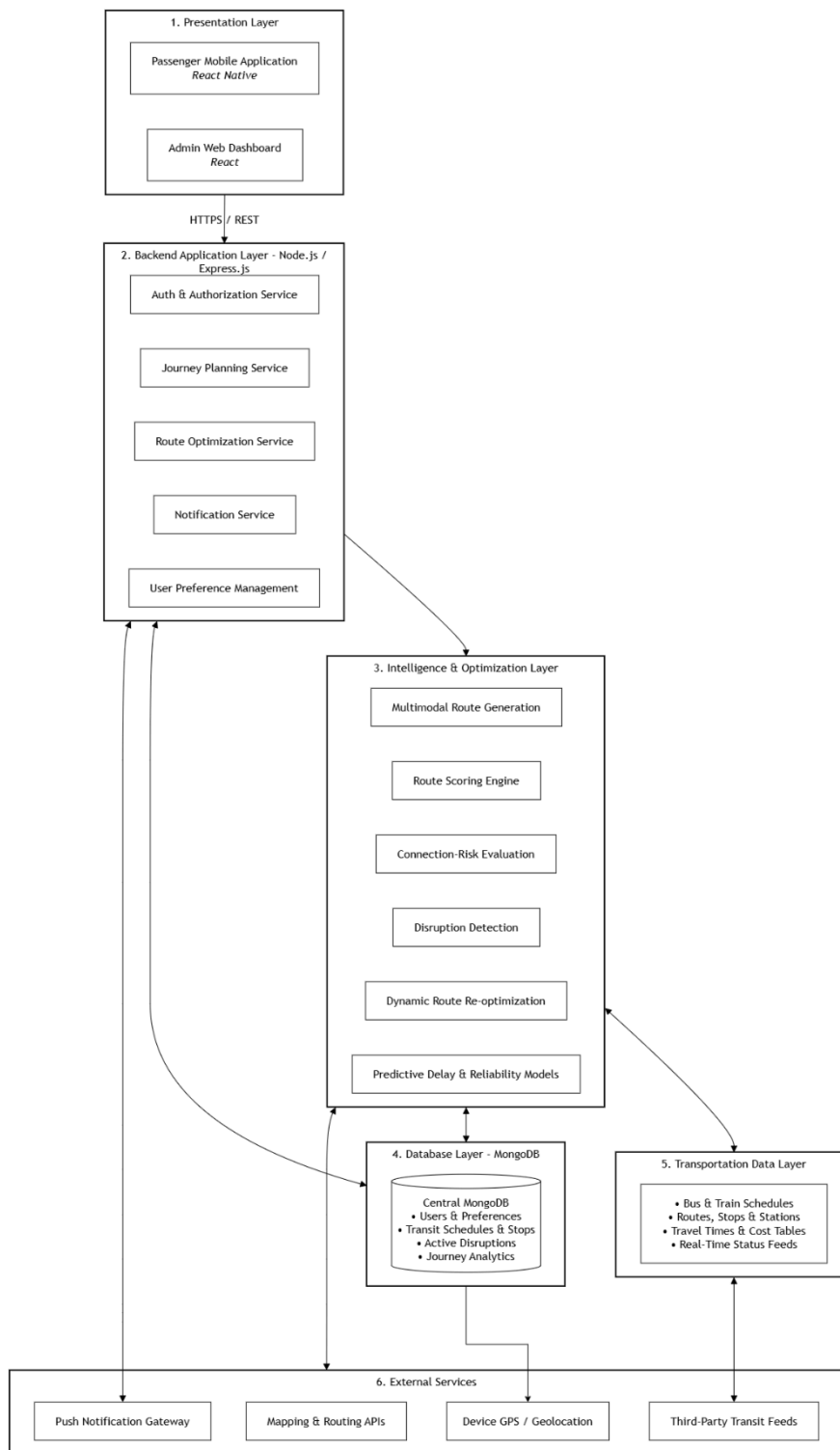


Figure 8.1: System Architecture

8.2 Multimodal Journey Data Flow

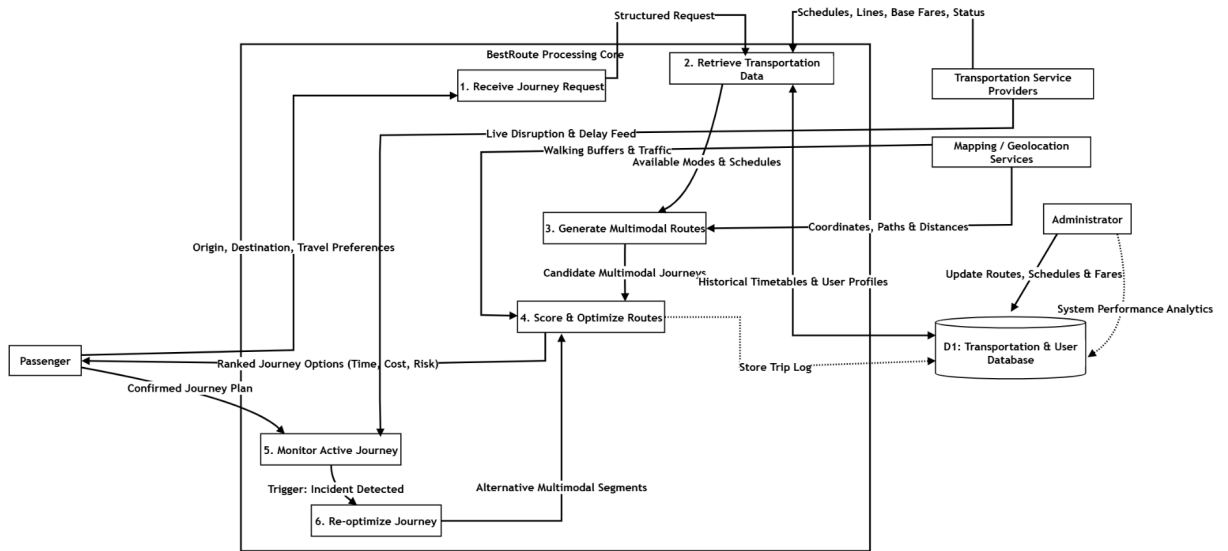


Figure 8.2: Data Flow diagram

8.3 Passenger Journey Planning Wireframe

From

↑↓

To

Travel Preference

Fastest

Cheapest

Search Journey ▼

Recommended Journeys

Recommended

Bus → Train → Taxi

Total time

Estimate fare

Transfers

Walking (m)

Risk

Cheapest

Route Map

Figure 8.3: Passenger Journey Planning Wireframe

8.4 Dynamic Re-routing Scenario

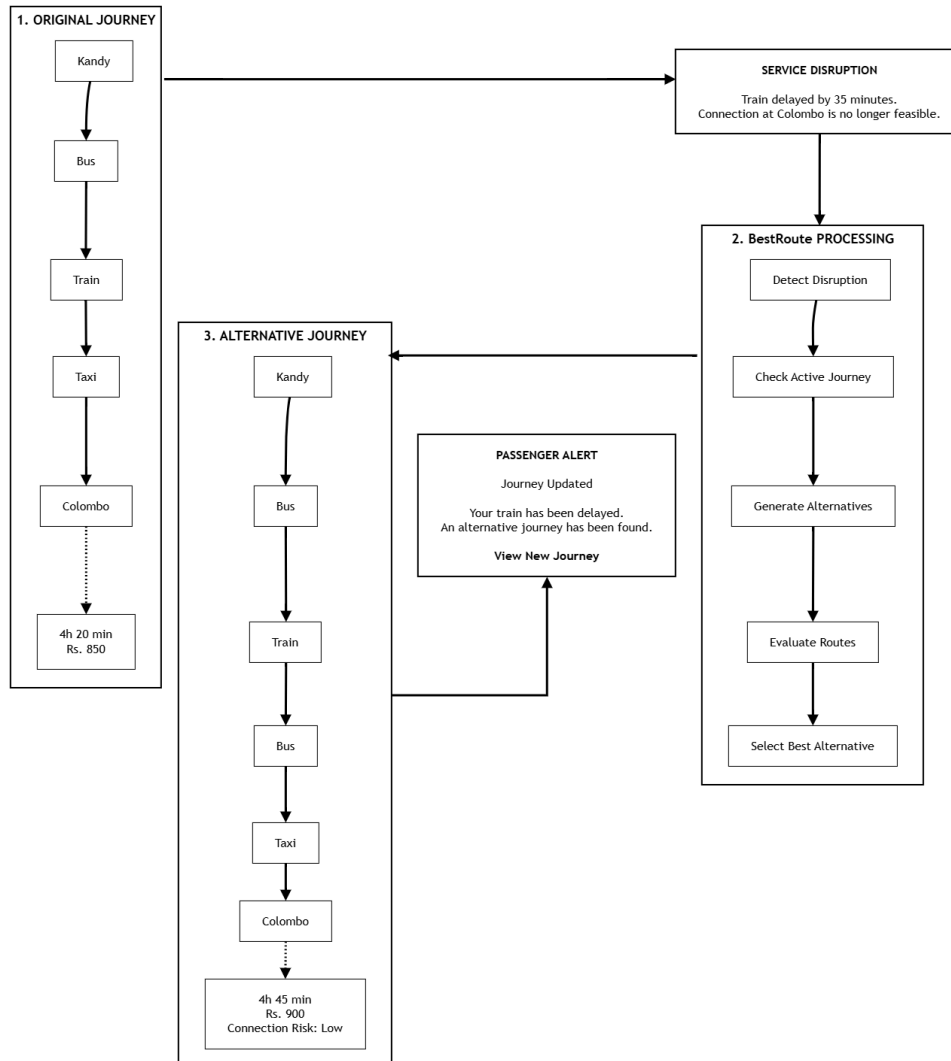


Figure 8.4: Dynamic Re-routing Scenario

8.5 User Flow

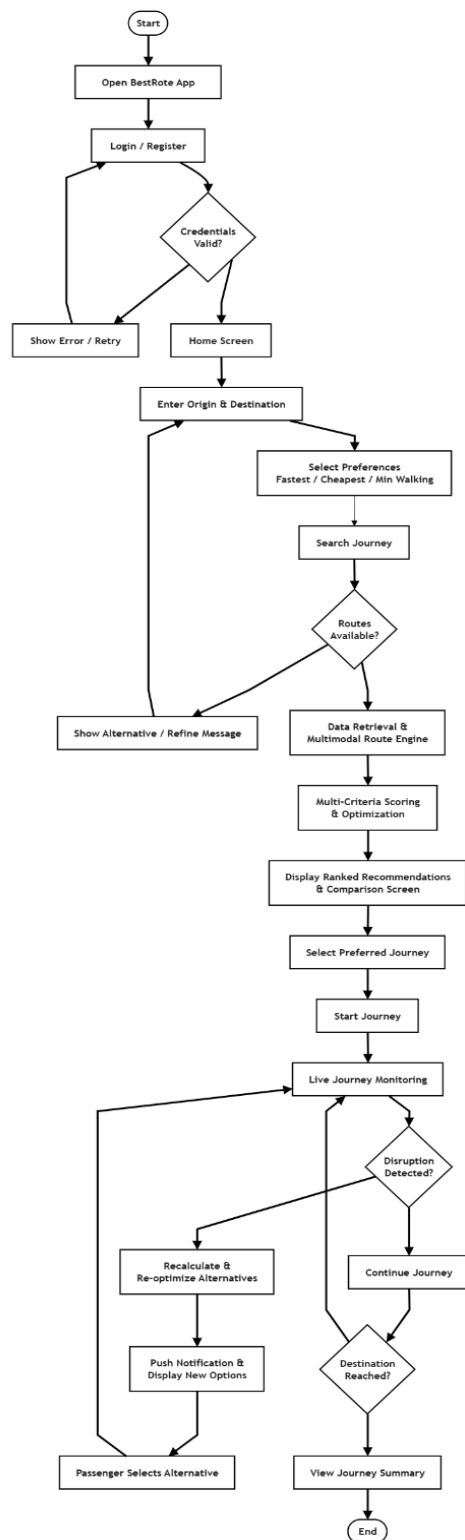


Figure 8.5: User flow

8.6 Use Case Diagram

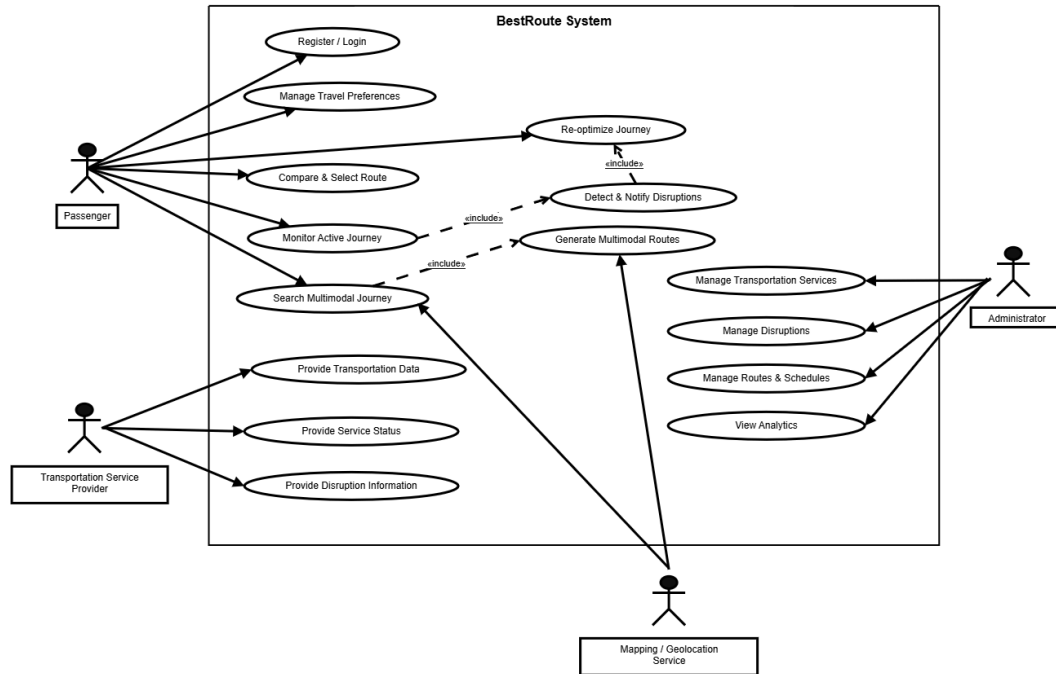


Figure 8.6: Use case diagram

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